Maryam Saeidmehr

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Profile

M.Sc. student in Artificial Intelligence at the University of Bologna and Erasmus+ exchange student at NTNU, interested in multimodal perception, visual foundation models, and embodied AI. Experienced in developing computer vision pipelines using transformer-based architectures, transfer learning, pseudo-labeling, and model ensembling. Seeking to contribute to research on robust visual perception and multimodal learning for real-world AI systems.

Education

NTNU – Norwegian University of Science and Technology — Trondheim, Norway
Erasmus+ Exchange – M.Sc. in Artificial Intelligence — *Aug. 2025 – July 2026*

Alma Mater Studiorum – Università di Bologna — Bologna, Italy
M.Sc. in Artificial Intelligence — *Sept. 2024 – Present*

Isfahan University of Technology — Isfahan, Iran
B.Sc. in Computer Engineering; ***GPA: 17.96/20 (3.84/4)*** — *Sept. 2017 – Feb. 2022*

Research Interests

Machine learning, Deep learning, Computer vision, Applied AI

Selected Projects

Snow Pole Detection for Autonomous Driving | *GitHub* — Spring 2026

- Developed a data-centric object detection pipeline for snow pole detection in Nordic winter driving conditions, where snow-covered roads make standard lane detection unreliable.
- Built a pseudo-labeling pipeline using Segment Anything Model (SAM) on scraped winter driving footage, expanding the training data beyond the original labeled dataset.
- Trained and compared YOLOv9t, YOLO11n, and RF-DETR models; combined CNN-based and transformer-based detectors using test-time augmentation and Weighted Boxes Fusion.
- Achieved 97.6% mAP@50 and 79.5% mAP@50:95 with the final ensemble, ranking first on the course project leaderboard.

COVID-19 Detector | *Undergraduate Research Assistant* | *GitHub* | *WebApp* — Oct. 2020 – Sept. 2021

- Implemented a deep convolutional neural network to detect Covid-19 from CT images using Keras and Tensorflow.
- Performed data pre-processing and augmentation using Python.
- Evaluated model performance using metrics such as accuracy, sensitivity, and specificity.
- Deployed model as a web application by using Django REST framework and ReactJS.

Image Watermarking in DCT Domain | *GitHub* — Fall 2019

- Designed and implemented an adaptive blind image watermarking algorithm concerning edge pixel concentration, improving imperceptibility by about 24% compared to non-adaptive methods.
- Reviewed NC results of both adaptive and non-adaptive methods against JPEG attack.

Skills

Programming: Proficient in C, C++, Python and Familiar with MATLAB

Machine Learning & Computer Vision: PyTorch, TensorFlow/Keras, Scikit-learn, NumPy, Pandas, OpenCV, YOLO, RF-DETR, SAM, transfer learning, pseudo-labeling, object detection, model ensembling

Web: HTML5, CSS3, Javascript, JQuery, ReactJS, Bootstrap, Django

Operating System: Linux (Ubuntu, Kali), Windows

Typesetting: Latex, Microsoft Office

Notable Courses

Computer Vision and Deep Learning, Robotic Vision, Visual Intelligence, Modern Machine Learning in Practice, Developing ML Tools: Insights and Case Studies, Artificial Intelligence, Fundamentals of Machine Learning, Recommender Systems, Applied Linear Algebra, Signals and Systems Analysis, Game Theory, Databases.

Research Experience

Isfahan University of Technology Isfahan, Iran
Supervisor: Prof. Shadrokh Samavi *Oct. 2020 – Sept. 2021*

- Proposed a deep convolutional neural network to efficiently detect COVID-19 from CT images using the transfer learning approach performing with a high degree of accuracy.

Teaching Experience

Isfahan University of Technology Isfahan, Iran
Teaching Assistant for the course Compiler *Feb. 2022 – June 2022*

- Assisted in creating course materials and procedures, grading, and conducting instructional workshops

Isfahan University of Technology Isfahan, Iran
Teaching Assistant for the course Databases I *Feb. 2021 – June 2021*

- Assisted in creating course materials and procedures, grading and conducting Q&A sessions

Work Experience

Freelance Isfahan, Iran
Full Stack Developer *June. 2023 – Oct. 2024*

- Built frontend pages for a smart-city platform (Municipality of Isfahan) covering air quality, transportation, and parking systems using React.js, Redux.js, and Tailwind CSS.
- Developed interactive data visualization interfaces for large-scale urban datasets using charts and mapping tools.
- Built an education-gamification web app combining homework management with a real-time multiplayer game using Django REST Framework, Channels, and WebSockets.
- Implemented responsive React.js and Material-UI frontend with real-time features enabling live student interaction.

Rasad Sarmayeh Company Isfahan, Iran
Data Analyst *June 2023 – Oct. 2023*

- Analyzed and interpreted large sets of marketing and trading data to inform business decisions.
- Cleaned and prepared data for analysis, ensuring accuracy and reliability.
- Designed and implemented SSIS packages to automatically extract, transform, and load data for efficient interpretation and analysis.
- Created visualizations and reports to effectively communicate data insights to stakeholders.

Zamin Company Isfahan, Iran
Software Engineer Intern *June 2021 – Sept. 2021*

- Assisted in building beautiful user interface for clients without compromising functionality for aesthetics.
- Learned front-end object-oriented programming to develop client server systems.
- Wrote unit tests in Jest to ensure code was tested and 100% bug free.

Awards & Achievements

Admitted to Erasmus+ Exchange Program with funding for second-year M.Sc. studies at NTNU, Norway — Aug. 2025

Ranked 10th Among More Than 80 Undergraduate Students in Computer Engineering Department, Isfahan University of Technology — Dec. 2017 – Feb. 2022

Ranked within the top 1% in 'National Entrance Exam for B.Sc Studies' in Iran Among More Than 148,000 Students in the Field of Mathematics and Physics — Aug. 2017

Admitted to NODET, National Organization for Development of Exceptional Talents — Aug. 2013